CUSTOMER CHURN PREDICTION – PROJECT DOCUMENTATION

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1. **Data Collection**

Objective:
Collect relevant data to understand customer behavior and predict churn in the telecom industry.

Description:

- The dataset was sourced from the **Telco Customer Churn dataset** (public Kaggle dataset).
- It contains **7,000+ customer records** with 21+ features, including:
    - **Demographics:** gender, senior citizen, partner, dependents
    - **Account information:** tenure, contract type, monthly charges, total charges
    - **Services:** phone, internet, security, backup, streaming
    - **Target variable:** Churn (Yes/No)

Methods:

- Data was downloaded as a CSV file.
- All files stored in data/raw/ for reproducibility.
- For future production, the data can be collected via **CRM databases or APIs**.

Key Points:

- No personal customer identification information is included.
- Ensured dataset has enough diversity for modeling high-risk vs loyal customers.

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2. **Data Preprocessing**

Objective:
Clean and prepare the raw data for machine learning models.

Steps Taken:

1. **Missing Values Handling**
    - Converted TotalCharges to numeric.

    - Dropped rows with missing values.

2. **Target Encoding**
    - Converted Churn column: Yes → 1, No → 0.
3. **Categorical Variables Encoding**
    - Used **one-hot encoding** for categorical features like Contract, PaymentMethod, tenure_group.
    - Dropped first dummy to avoid multicollinearity.
4. **Handling Imbalanced Dataset**
    - Original churn data is **imbalanced (~26% churn, 74% stay)**.
    - Used **SMOTE** (Synthetic Minority Oversampling Technique) to balance classes for training.
5. **Feature Scaling**
    - Applied **StandardScaler** for numerical features to normalize values, improving model convergence.

Outcome:

- Clean dataset with no missing values, properly encoded, balanced target classes.
- Ready for feature engineering and modeling.

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3. Univariate and Bivariate Analysis

Objective:
Understand distributions of individual features (univariate) and relationships between features and the target (bivariate).

Univariate Analysis

- Plotted histograms for continuous features (tenure, MonthlyCharges, TotalCharges).
- Observations:
    - Most customers have **tenure < 30 months**.
    - MonthlyCharges is skewed; higher charges correlate with churn.
- Plotted countplots for categorical features (Contract, InternetService, gender).
    - Observed: **Month-to-month contracts churn more**.

Bivariate Analysis

- Plotted **Churn vs Tenure** using boxplots:
  - Low tenure → higher churn.
- **Churn vs Monthly Charges**:
  - High monthly charges → higher churn.
- Correlation heatmap for numeric features:
  - tenure negatively correlated with churn.
  - MonthlyCharges positively correlated with churn.

Outcome:

- Identified **features with predictive power** for churn.
- Found patterns for feature engineering.

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4. **Feature Selection and Model Creation**

Objective:
Select the most informative features and build machine learning models for churn prediction.

Feature Engineering

- Tenure groups: Low, Medium, High
- Avg Monthly Spend = TotalCharges / (Tenure + 1)
- Service Count = sum of active services
- Engagement Score = Tenure × Service Count

Feature Selection

- Removed low-variance features or redundant columns.
- Used **correlation and domain knowledge** to select final features.
- Features used for modeling:
  - Tenure, MonthlyCharges, TotalCharges, ServiceCount, EngagementScore, Contract type, Internet service, tenure group, etc.

Models Created

- Logistic Regression

- Random Forest
- XGBoost (Optuna tuned)
- LightGBM
- SVM
- KNN
- MLP Neural Network

Model Training

- Train-test split (80-20)
- Balanced training data using SMOTE
- Scaled numeric features with StandardScaler

Model Comparison Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Ensemble Models:

- Stacking: Random Forest + XGBoost + LightGBM, final estimator = Logistic Regression
- Voting Classifier (soft voting)

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5. **Final Model**

Best Model:

- **Stacked Ensemble (Random Forest + XGBoost + LightGBM)**
- **Hyperparameters tuned using Optuna**

Evaluation on Test Set:

- Accuracy: ~0.85
- ROC-AUC: ~0.90
- Confusion Matrix: Balanced prediction between churn and stay classes

Explainability:

- Used **SHAP values** to explain predictions:
  - High Monthly Charges → increases churn probability
  - Low tenure → increases churn probability
  - Multiple services → reduces churn probability

Customer Segmentation (KMeans):

- 3 Segments:
  - High-risk customers
  - Loyal customers
  - New customers
- Segments used for targeted retention strategies

Business Insight Layer:

- Predicted churn + reasons + segment → actionable recommendations:
  - "High charges → offer discount"
  - "Low tenure → onboarding program"

Deployment:

- Streamlit app built for interactive predictions and visualization:
  - Input features → predict churn → explain top reasons → show customer segment

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6. Summary

This project demonstrates a **full ML pipeline for churn prediction**, from raw data collection to **deployment-ready application**.

Key Features:

- Advanced feature engineering (tenure group, engagement, service count)
- Model diversity + hyperparameter tuning (Optuna)
- Ensemble learning (Stacking)
- Explainable AI (SHAP)
- Customer segmentation (KMeans)

- Business actionable insights

Impact:

- Predicts churn **accurately**
- Explains **why customers churn**
- Provides actionable **retention strategies**
- Ready for real-world deployment